

Lecture 2 Introduction to Linux Notes

1. What is an Operating System?

- An operating system is what allows users all the fundamental software features, such as using the computers hardware, and providing all the basic tools that makes a computer useful.

2. What is a Kernel?

- A kernel is the component which is responsible for the low level features such as managing the hardware, memory allocation, cpu time, and how programs interact with one another.

3. Which other parts aside from the kernel identify an OS?

1. Command-Line Shells
2. Graphical User Interfaces
3. Utility and Productivity Programs
4. Libraries

4. What is Linux?

- Linux is a an operating system that consists of kernel, libraries, and utilities that makes up the operating system.

5. What is a Linux distribution?

- A linux distribution is any kind of operating system that runs the linux kernel like core unix tools, startup scripts, supplemental software.

6. List at least 4 Linux characteristics:

1. Open source
2. User interface
3. Cost is typically free
4. Configurability

7. What is Debian?

- Debian is known as the grandfather of all linux, its been dedicated to create free software while promoting the idea of free software to a community.

8. List and define the different types of licensing agreements

1. *Open source*
 - the software is distributed either with a fee or for free.
2. *Closed source*
 - unlike open source the software is not distributed with the source code, and the user is restricted from making any changed the code.

3. Freeware

- the software is free but the source code is no longer available.

4. Shareware

- Unlike freeware, shareware is also free with a trial basis.

9. What is Free software? Define the 4 freedoms.

- Free software is a software that has been shared via source code, it can either be handed out for free or it could be obtained with a fee.

1. Freedom 0

- Uses software for any reason.

2. Freedom 1

- Examines the code and makes any changes as you wish

3. Freedom 2

- The software can be redistributed.

4. Freedom 3

- Software that's been modified can be redistributed.

10. What is virtualization?

- Virtualization is a space that is virtually created, in the case of computers, its often used to run multiple operating systems on one machine. Essentially dividing the hardware to create multiple computers inside of on computer.